Assignment 02 Report

React Native Instagram Feed & Profile UI Development

Summer 2026 Internship

SUBMITTED BY — Ms. Marjan Yousafzai

TN/IN03/MD/001

Date: 07/23/2026

SUBMITTED TO — Mr. Muhammad Ali

Mobile App Development

Technik Nest Pvt Ltd: https://techniknest.com/

LinkedIn: https://pk.linkedin.com/company/techniknest

1. Objective

The objective of this assignment was to develop two Instagram-inspired user interfaces using React Native:

- Instagram Feed Screen with stories and posts.
- Instagram Profile Screen with user information, statistics, story highlights, and photo grid.

The purpose of this project was to strengthen the understanding of React Native components, Flexbox layouts, lists, reusable components, image handling, and responsive mobile UI design.

The assignment focused solely on the user interface (UI) and did not require backend development, authentication, database integration, or API integration. All application content was displayed using dummy data and locally stored images.

2. Requirements

2.1 General Requirements

- Design and develop both Feed and Profile screens.
- Use React Native core components.
- Implement responsive layouts.
- Use dummy data for application content.
- Use local images for stories, posts, profile images, and grid images.
- No backend or API integration.
- Maintain a clean and organized project structure.
- Use reusable components where appropriate.

- Implement user interactions for selected UI elements.

2.2 Feed Screen Requirements

Header

- Instagram logo or text.
- Heart or notification icon.
- Messenger or chat icon.
- Header positioned at the top of the screen.

Stories Section

- Horizontally scrollable stories section.
- Circular profile pictures.
- Username displayed below each story.
- Minimum of 10 stories.
- First story represents "Your Story."
- Plus icon displayed on the "Your Story" profile image.
- Equal and consistent spacing between stories.

Posts Section

Each post should include:

- User profile picture.
- Username.
- Location where applicable.
- Three-dot menu icon.
- Large responsive post image.
- Like, comment, share, and save icons.

- Number of likes.
- Post caption.
- Comments count.
- Time posted.
- Minimum of 5 posts.

2.3 Profile Screen Requirements

Profile Header

- Circular profile picture.
- Username.
- Full name.
- Bio.
- Website link displayed as text.

Statistics Section

- Posts count.
- Followers count.
- Following count.

Profile Buttons

- Edit Profile.
- Share Profile.
- Contact.
- Rounded corners.
- Equal or consistent button widths.
- Consistent spacing.

Story Highlights

- Horizontally scrollable highlights.
- Circular highlight images.
- Highlight titles.
- Minimum of 5 highlights.

Profile Tabs

- Posts.
- Reels.
- Tagged.
- User interface only without additional functionality.

Posts Grid

- Three-column grid layout.
- Minimum of 12 images.
- Equal spacing between images.
- Square image format.
- Efficient list-based grid layout.

3. Tools and Technologies Used

The following tools and technologies were used to develop the application:

Category	Technology
Framework	React Native
Development Platform	Expo
Library	React

Category	Technology
Programming Language	JavaScript with JSX
Navigation	React Navigation
Styling	React Native StyleSheet
State Management	React Hooks
Animations	React Native Animated API
List Rendering	FlatList
Responsive Design	Dimensions API
Development Environment	Visual Studio Code

Key Libraries Used

- **React Navigation** - Used for screen and navigation management.
- **React Navigation Bottom Tabs** - Used for bottom navigation between application screens.
- **React Native Screens** - Used for screen handling and navigation.
- **React Native Safe Area Context** - Used for handling device safe areas.

4. Project Folder Structure

The project follows an organized folder structure to separate application files, assets, configuration, and documentation.

Project Name: Instagram

- **app**

    - **index.jsx** - Main application file containing the Feed and Profile UI.
- **assets**
    - **images** - Contains local images used throughout the application.
        - Story images.
        - Post images.
        - Profile images.
        - Grid images.
        - Story highlight images.
- **package.json** - Contains project dependencies and configuration.
- **app.json** - Contains Expo application configuration.
- **.gitignore** - Specifies files and folders excluded from Git.
- **README.md** - Contains project documentation and setup instructions.

5. Application Features

5.1 Feed Screen Features

- Instagram-inspired header with logo and action icons.
- Horizontally scrollable stories section.
- Circular story profile pictures.
- "Your Story" with a plus icon.
- 10 or more stories with usernames.
- Multiple posts with images and captions.
- Interactive like and unlike functionality.
- Save and unsave functionality.

- Like counter with dynamic updates.
- Comments count display.
- Post time display.
- Instagram-inspired post layout.

5.2 Profile Screen Features

- Circular profile image.
- Username and full name display.
- Multi-line profile bio.
- Website link display.
- Profile statistics including Posts, Followers, and Following.
- Three profile action buttons.
- Horizontally scrollable story highlights.
- Highlight titles and images.
- Three profile tabs including Posts, Reels, and Tagged.
- Three-column photo grid.
- Minimum of 12 profile grid images.

5.3 Interactive Features

- Full-screen story viewer.
- Story viewer animations.
- Story progress bar.
- Like and unlike functionality.
- Dynamic like counter updates.
- Save and unsave functionality.

- Bottom navigation between Feed and Profile screens.
- Smooth animations and transitions.
- Interactive touch-based UI elements.

6. Working Explanation

6.1 Application Overview

The application is divided into two main sections: the Feed Screen and the Profile Screen. The Feed Screen provides an Instagram-inspired home page where users can view stories and posts. The Profile Screen displays user information, statistics, story highlights, navigation tabs, and a photo grid.

The application uses dummy data and local image assets to create a realistic interface without requiring a backend or external API.

6.2 Feed Screen Working

The Feed Screen contains:

- Application header.
- Stories section.
- Posts section.
- Post interaction buttons.

The Stories section displays multiple user stories in a horizontal scrolling layout. Each story contains a circular profile image and a username.

The first story is presented as "Your Story" and includes a plus icon.

Below the Stories section, multiple posts are displayed vertically. Each post contains a profile header, post image, action buttons, likes, caption, comments, and posting time.

The Like button allows users to increase or decrease the displayed like count. The Save button provides an interaction for saving and unsaving posts.

6.3 Profile Screen Working

The Profile Screen presents an Instagram-inspired user profile.

The top section displays:

- Profile picture.
- Username.
- Full name.
- Biography.
- Website information.
- Posts count.
- Followers count.
- Following count.

Below the profile information, users can see profile action buttons such as Edit Profile, Share Profile, and Contact.

The Story Highlights section displays circular images with titles in a horizontally scrollable layout.

The Profile Tabs section contains Posts, Reels, and Tagged options.

The Posts section displays profile images in a three-column square grid.

6.4 Story Viewer

The application includes a full-screen story viewer for displaying selected stories.

The Story Viewer includes:

- Full-screen story image.

- Progress indicator.
- Opening and closing animations.
- Automatic story progress.
- Option to close the story viewer.

6.5 Responsive Design

The application was designed to provide a responsive user interface across different mobile screen sizes.

Responsive design techniques were used for:

- Post image sizing.
- Profile grid dimensions.
- Story image sizing.
- Screen spacing.
- Button layouts.
- Header alignment.

6.6 Reusable Components

The application uses reusable components to maintain clean and organized code.

The main reusable components include:

- **StoryItem** - Displays an individual story.
- **PostCard** - Displays an individual post.
- **HighlightItem** - Displays a story highlight.
- **GridItem** - Displays an individual profile grid image.
- **StoryView** - Displays the full-screen story viewer.

7. How to Run the Project

7.1 Prerequisites

- Node.js.
- npm or Yarn.
- Expo development tools.
- Visual Studio Code.
- Android Emulator or physical Android device.
- iOS Simulator for macOS users.
- Expo Go application for physical device testing.

7.2 Installation Steps

Step 1: Clone the Repository

Clone the project repository from GitHub and open the project folder.

Step 2: Install Dependencies

Install all required project dependencies.

Step 3: Start the Development Server

Start the Expo development server from the project directory.

Step 4: Run the Application

The application can be opened using:

- Android Emulator.
- iOS Simulator.
- Web browser.
- Physical Android or iOS device using Expo Go.

Step 5: Test the Application

Test the following features:

- Feed Screen.
- Stories section.
- Post interactions.
- Story viewer.
- Profile Screen.
- Story Highlights.
- Profile grid.
- Bottom navigation.

8. Useful Terminal Commands

The following terminal commands were used during development and testing:

- **cd instagram-ui-clone** - Opens the project directory.
- **npm install** - Installs project dependencies.
- **npx expo start** - Starts the Expo development server.
- **npx expo start --web** - Runs the application in a web browser.
- **npx expo start --tunnel** - Starts Expo using Tunnel mode.
- **npx expo start --ios** - Opens the project in the iOS Simulator.
- **npx expo start --android** - Opens the project in the Android Emulator.
- **r** - Reloads the application.
- **w** - Opens the application in a web browser.
- **a** - Opens the application in an Android Emulator.
- **i** - Opens the application in an iOS Simulator.
- **Ctrl + C** - Stops the Expo development server.

9. Screenshots

Source Code

The index.jsx file contains the main application code responsible for implementing the Feed Screen, Profile Screen, reusable UI components, interactive functionality, and navigation.

index.jsx

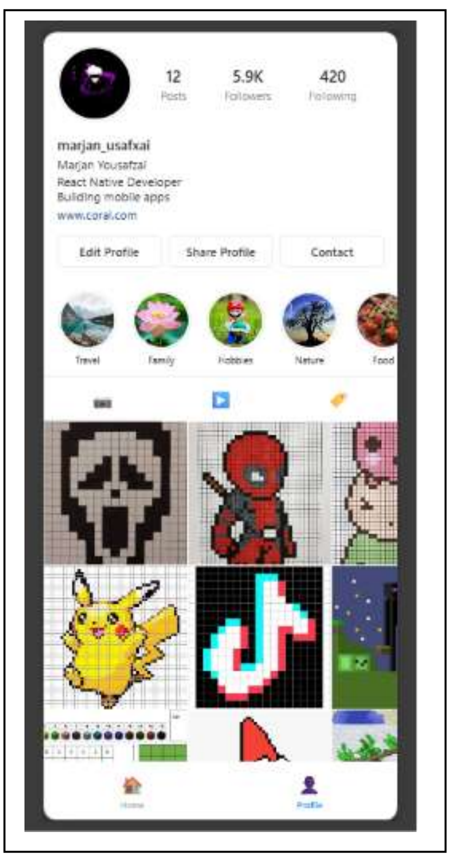
https://github.com/yousafzai05/InstagramUI/blob/main/app/index.jsx

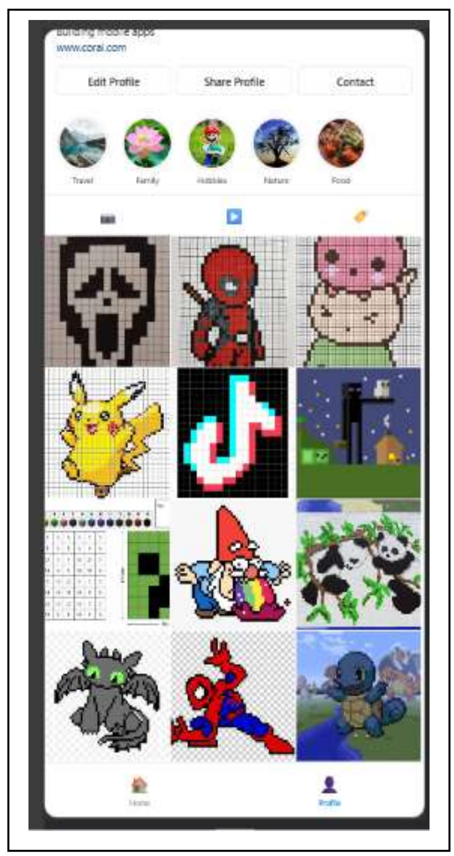
Application Output

Feed Screen

Story Viewer

Profile screen





10. Links

GitHub Profile

https://github.com/yousafzai05

GitHub Repository

https://github.com/yousafzai05/InstagramUI

LinkedIn Profile

https://linkedin.com/in/marjan-yousafzai

11. Conclusion

This assignment successfully developed two Instagram-inspired user interfaces using React Native and Expo.

The Feed Screen includes an Instagram-inspired header, a horizontally scrollable Stories section with 10 or more stories, and multiple posts with images, captions, likes, comments, and interactive functionality.

The Profile Screen includes a complete user profile with profile statistics, action buttons, Story Highlights, navigation tabs, and a three-column photo grid containing 12 or more images.

The project demonstrates practical knowledge of React Native components, state management, responsive layouts, image handling, animations, navigation, and user interaction.

The assignment strengthened practical skills in mobile application UI development and provided hands-on experience with React Native and Expo development workflows.

The completed application provides a realistic Instagram-inspired user interface while fulfilling the required Feed Screen and Profile Screen specifications. The application uses dummy data and local images without requiring backend, database, authentication, or API integration.